SkillNavora · PM Master Handbook

MODULE 7 · DETAILED CHAPTER

Product Analytics and Metrics

How product managers define success, track it every day, and turn numbers into decisions.

Based on the Week 6 Analytics and Metrics class (Product Management Launchpad). Written by Varun Kumar.

Part 1 · How PMs define success

1. What a metric is
2. Why metrics matter
3. From user value to business value
4. The mental model for product metrics
5. Exercise: Instagram Reels
6. The ladder of metrics
7. Leading vs lagging, input vs output
8. Build a metric system, not one metric
9. B2B vs B2C metrics
10. B2B health dashboard and funnels
11. Worked activity: Google Maps
12. Goodhart's Law and metric gaming
13. Counter-metrics and guardrails

Part 2 · How PMs track metrics day to day

14. AARRR framework
15. Retention and product-market fit
16. Growth loops beyond the funnel
17. HEART framework
18. OKRs and the metric cascade
19. Analytics tools hierarchy
20. Metrics cheat sheet
21. Interview answer template

PART 1 · HOW PRODUCT MANAGERS DEFINE SUCCESS

1. What a metric is

A metric is a quantifiable measure used to assess success, or failure, against a goal. The metrics that give the most insight are called KPIs (Key Performance Indicators).

Anatomy of a metric

Part	What it means	Example
Definition	What the metric tries to measure	Count of users, depth of engagement, degree of retention
Value	The numerical value	12,400 weekly active users
Time grain	The period the measurement covers	Monthly vs weekly active users; week-4 vs week-on-week retention
Cohort	The slice of users being measured	New users vs existing users

A metric without a time grain and a cohort is ambiguous. "Retention is 30%" means little until you know which users, measured when.

QUOTE TO REMEMBER

"You can't manage what you don't measure." (Peter F. Drucker)

2. Why metrics matter

Metrics turn product work into clear decisions. They do four jobs in a loop:

Track See what is happening.

Diagnose Understand why it happened.

Decide Choose what to improve.

Align Show progress to the team.

Without metrics, opinions dominate. With metrics, patterns become visible. The good PM habit is simple: **measure before you change**.

3. From user value to business value

Good product metrics connect what users are trying to achieve with what the business is trying to grow. Think of it as a bridge with three steps:

1. **User job:** the outcome the user wants (for example, "get set up and go live faster").
2. **User value metric:** a measure that the user job is working (for example, time to first completed workflow).
3. **Business value metric:** a measure of business impact (for example, activation rate, expansion potential, lower churn risk, revenue).

Example: a B2B onboarding platform. Faster setup leads to a shorter time-to-value, which leads to renewal confidence.

PART 1 · HOW PRODUCT MANAGERS DEFINE SUCCESS

4. The mental model for product metrics

The goal of every product is to **solve customer problems** and **generate value for the business**. Metrics must cover both sides.

CUSTOMER SIDE

1. **Identify all users and stakeholders.** One product can have more than one type of user.
2. **Identify their Jobs To Be Done (JTBD).** The higher-order problem the user wants solved.
3. **Define a success metric** that captures completion of the JTBD. Often one or more metrics capture true success for the user.

BUSINESS SIDE

1. **Identify the business goal.** For-profit companies ultimately aim to generate shareholder value. Vision and mission statements also make excellent goal statements.
2. **Define a success metric** that shows the product supports the business model.
3. **Add guardrail (counter) metrics** to guard against side effects.

Worked example: a goal that is hard to measure

The class example starts with "to lead a successful life". The metrics are a set, not one number:

Area	Metric
Financial health	Net worth
Physical health	Set of biomarkers (for example HbA1c, lipid panel)
Mental health	Cortisol level
Social health	Number of meaningful relationships

Two lessons follow. Defining success is hard when the goal is ambiguous. And you usually need many metrics to capture success.

5. Exercise: Instagram Reels

Work through the three stakeholders of a short-video feature:

Stakeholder	JTBD	Success metric (JTBD completion)
Content creators and influencers	Build a fan following or a captive audience	Total user reach, total followers, total creates
Content consumers	Find inspiring, relatable or funny content	Total watch time, total likes, comments and shares
Advertisers and brands	Influence targeted users to buy	Total ad impressions, clicks, purchases

The business-goal metrics for the same product: **revenue**, **number of new users added**, and a **guardrail** such as a drop in overall use of the app.

PART 1 · HOW PRODUCT MANAGERS DEFINE SUCCESS

6. The ladder of metrics

Metrics come in categories. Knowing the category tells you what you can control, when you will see the signal, and how it connects to the result.

North Star Metric (NSM)

The single metric, sometimes called the One Metric That Matters (OMTM), that signifies that the product is delivering value. Examples from the class:

- **Swiggy:** number of successful orders delivered.

- **Airbnb:** number of bookings (and successful stays).
- **Spotify:** total listen time, supported by a "magic moment" (users who follow 10 artists in the first week).

Build the ladder under the North Star

The Airbnb example in the class builds a ladder under its North Star:

Rung	Type	Airbnb example
North Star	The one action that creates value	Number of bookings (and successful stays)
Input	What we control	Number of high-quality listings added
Output	What we get	Number of bookings made
Leading	Early signal, predicts the future	Number of views on listings
Lagging	Confirms the past	Revenue, e.g. commission per booking

Spotify in practice

Leading metrics (# of app installs, monthly active users, # of playlist creations, efficacy of the recommendation system) feed the North Star of total listen time. The magic moment (following 10 artists in the first week) sits upstream of listening. Lagging metrics (subscription revenue, ad revenue, other revenue) confirm the outcome.

THE RULE OF THUMB

If inputs improve, outputs usually follow. Leading metrics tell you where you are going. Lagging metrics tell you where you reached.

Input and output, in one line each

- **Input metrics** measure the controllables you can directly influence, such as the number of user invites sent.
- **Output metrics** measure the outcomes that inputs cause, such as total sign-ups from user invitations.

PART 1 · HOW PRODUCT MANAGERS DEFINE SUCCESS

7. Leading vs lagging, input vs output: the 2×2

Put the two axes together. Controllability (can the team directly move it?) and temporality (does it predict the future or report the past?) give four quadrants.

LEADING INPUTS (HIGH CONTROL, PREDICTIVE)

The daily levers product and growth teams control. Slack: number of team-member invites sent during onboarding. SaaS CRM: active CRM integrations connected on Day 0. E-commerce: autocomplete latency and zero-result query rate.

LEADING OUTPUTS (LOW CONTROL, PREDICTIVE)

Early customer behaviour that signals long-term value. Slack: a team reaching 2,000 sent messages (the magic moment). Spotify: user saving 10+ songs and following 3 artists in week 1. Duolingo: day-7 streak completion rate.

LAGGING INPUTS (HIGH CONTROL, RETROSPECTIVE)

Actions the organisation completed in the past. Engineering: story points shipped, PRs merged, sprint velocity. Growth: total paid marketing budget deployed. Content: help-centre articles

LAGGING OUTPUTS (LOW CONTROL, RETROSPECTIVE)

The business and financial outcomes. Annual recurring revenue (ARR), net dollar retention (NDR), customer lifetime value (LTV), LTV:CAC

published this quarter. These measure effort, not customer value.

ratio, quarterly EBITDA. These are the scoreboard. You cannot operate them directly.

8. Build a metric system, not one metric

A single metric rarely tells the full story. A complete system has five layers. Example: a B2B collaboration product.

Layer	Question it answers	Example
North Star	What long-term value are we creating?	Weekly active teams
Input metrics	What behaviours drive the result?	Invites sent, workflows created
Output metrics	What moved after launch?	Activation rate, workflow completion
Guardrails	What should not get worse?	Error rate, latency
Diagnostic metrics	Where exactly is the problem?	Drop-off after the invite step

PART 1 · HOW PRODUCT MANAGERS DEFINE SUCCESS

9. B2B vs B2C metrics

A product can look similar in both models, but the metric logic is often different.

	B2C (consumer)	B2B (business)
Unit of behaviour	User-level behaviour	Account or team-level behaviour
Decision	Shorter decision cycle	Multi-stakeholder adoption, longer sales and renewal cycle
Focus	Conversion and retention	Activation by account, seat adoption, churn risk, expansion revenue
Example metrics	DAU, checkout conversion, repeat rate	Account activation rate, seats adopted, renewal health

10. B2B health dashboard and funnel analysis

A B2B PM needs a dashboard that shows **value**, **risk** and **growth** at the account level:

Lens	Question	Metrics on the dashboard
Value	Are accounts getting live and using the product?	Activation rate, weekly active accounts, adoption breadth, time-to-value
Risk	Which accounts are slipping, blocked or disengaging?	Renewal health, health score distribution, top at-risk accounts
Growth	Which accounts are ready to expand?	Expansion potential, expansion revenue

Funnel for B2B activation

A funnel shows where accounts drop before they become active. The class example:

Step	Share of accounts	What the drop usually means
1. Account created	100%	Starting point
2. Workspace connected	74%	Setup friction
3. First teammate invited	48%	Collaboration value unclear
4. First workflow completed	31%	Product complexity too high
5. Weekly active team	22%	Habit not formed yet

Funnel analysis answers three questions: where users drop, which step blocks value, and what to test next. The biggest leak is usually the best experiment to run.

PART 1 · HOW PRODUCT MANAGERS DEFINE SUCCESS

11. Worked activity: metrics for Google Maps

The class asked students to do three things: identify the users and their JTBD, define a metric for JTBD completion and a metric for business support, then build a metric ladder with leading, lagging, input and output relationships.

	Drivers and commuters	Local businesses	Google platform
Core JTBD	Find the fastest, safest, congestion-free route with a reliable arrival ETA	Turn nearby foot traffic into verified paying customers	Build the world's most trusted spatial map and monetise high-intent commercial search sustainably
North Star	% of trips taken on optimal routes (time, road condition, traffic, tolls)	Place-sheet actions: calls, direction clicks, table reservations, menu views	Net high-intent monetisation (local search ads plus Maps platform API revenue)
Leading	# of active navigation sessions started	# of claimed business profiles, updated menus	Sponsored pin impressions, API query volume
Lagging	Weekly active drivers, 90-day app retention	Merchant ad spend, quarterly merchant retention	Consolidated ads and enterprise recurring revenue
Input → output	Reroute latency (under 2 s), hazard reporting speed → % completed optimal trips, trip NPS	Profile completeness, review response rate → place-sheet views, in-store visits	Sponsored pin context relevance → ad click-through rate, enterprise renewals
Guardrails	Trip abandonment rate (under 2%), ETA error (under 3 min), distracted-driving alerts	Fake or duplicate listing rate (under 0.1%), spam review volume, dispute resolution time	Map clutter index, navigation task completion drops, battery drain, privacy opt-outs

WHY THIS MATTERS

Each stakeholder gets its own JTBD and metric, but all of them connect to one business metric. The guardrails are what stop one side from winning at the other's expense. For example, pushing more sponsored pins could hurt the driver's trip NPS.

PART 1 · HOW PRODUCT MANAGERS DEFINE SUCCESS

12. Goodhart's Law and metric gaming

THE PRINCIPLE

"When a measure becomes a target, it ceases to be a good measure." (Charles Goodhart, economist. Marilyn Strathern's later formulation is the version most often quoted.)

How metrics get gamed

- **Proxy displacement.** Teams confuse the proxy (clicks, raw DAU) with the real value (completing the core workflow).

- **Path of least resistance.** Under sprint or OKR pressure, teams optimise the easiest way to move the number, not the hardest way to create value.
- **Compounding side-effects.** Unmonitored dimensions degrade silently until trust, retention or unit economics collapse.

Real PM traps from the class

Trap	What happens
Notification trap	Sending five promotional push alerts boosts DAU by 8% in the short term. App uninstalls and muted alerts then rise sharply.
Signup friction trap	Pre-checking a newsletter box lifts registration conversion. It produces low-intent churn and spam complaints.
Feature tour trap	Forcing five unskippable walkthroughs claims 100% feature adoption. Users click "Next" without learning anything.

The senior PM defence

- **Paired counter-metrics.** Never judge an output metric alone. Pair volume with quality, retention and friction guardrails.
- **Holdout groups.** Keep a permanent holdout (for example 5% of users) to prove true incrementality against what would have happened anyway.
- **Qualitative triangulation.** Pair the dashboard with weekly customer interviews and support tickets to catch metric distortion early.

13. Counter-metrics and guardrails

Every primary metric should be balanced by a guardrail that protects the user or the platform. The class architecture:

Focus area	Primary metric (what we push)	Gaming risk (what can break)	Counter guardrail (what we defend)
User acquisition	New sign-ups and app installs; blended and paid CAC	Low-intent click-farm traffic, bot sign-ups, deceptive ad conversions	Day-30 retention at or above 25%; chargeback and fraud rate under 0.1%
Engagement and growth	DAU; push alert open rate and CTR	Over-notifying users to create fake DAU spikes; annoyance, muting, churn	App uninstalls per 1K sessions under 1.5%; notification mute or opt-out rate
Monetisation and pricing	ARPU; paywall conversion and upsell rate	Dark patterns at checkout, hidden auto-renewals, aggressive paywalls	Support-ticket ratio under 2%; 30-day refund and dispute volume
Platform reliability	Feature release velocity; experiment concurrency	Shipping unvalidated code, bloated dependencies, latency	P99 end-to-end latency under 400 ms; crash-free sessions at or above 99.95%

SENIOR PM RULE

Never ship an experiment without its paired guardrail. If the primary conversion rate rises 5% but a guardrail trips (churn or support tickets surge), the experiment is rolled back automatically.

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

14. The AARRR framework

AARRR (Dave McClure's "pirate metrics") walks through the customer lifecycle in five stages. Each stage asks one question.

Stage	Question it asks	Sample metrics
Acquisition	Are users showing interest in our product?	App downloads, site visits, social media impressions, CAC
Activation	Are users using our product (reaching the magic moment)?	Active users, % reaching the first value event
Retention	Are users using our product repeatedly?	Daily, weekly or monthly retention; churn rate
Referral	Are users bringing in more users?	Net Promoter Score (NPS), viral coefficient, referral share
Revenue	Are users generating revenue for us?	Sales, lifetime value (LTV)

What each stage teaches

- **Acquisition** happens when a customer is acquired through ads, SEO, content, sales, affiliates, community or referrals. Customers move through the AIDA funnel (awareness, interest, desire, action). High acquisition with poor retention is a leaky bucket. Growth must be steady because some users always leave.
- **Activation** happens when a customer has used the product and started deriving value. Products deliberately push users to the magic moment. Twitter suggested popular accounts after users followed 30 people. Dropbox encouraged users to upload a file during sign-up. If acquisition is broad and untargeted, most users never activate.
- **Retention** requires the customer to repeat the key value-driving action over time. Habit-forming products retain best. Users who churned can often be reactivated.
- **Referral** happens when a customer recommends the product. Organic referral chains keep acquisition costs low. Products create viral loops through referral discounts, contact-list invites and shareable content.
- **Revenue** depends on the right business model. Lifetime value must be much greater than customer acquisition cost (LTV ≫ CAC) for the business to be sustainable.

AARRR applied: Cred (credit card bill payments)

Stage	Metric
Acquisition	# of app installs
Activation	# of users who pay their first credit card bill, or set up their cards on Cred
Retention	# of users who pay n consecutive credit card bills
Referral	# of users who share rewards on social media, or sign up with a referral code or link
Revenue	# of users who buy products or services using Cred rewards

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

15. Retention and product-market fit

Long-term retention is one of the best signs of product-market fit (PMF). Retention curves make it visible.

Two products are shown on the same axes over 20 time units. **Product A** drops sharply at first, then flattens into a stable tail. It has product-market fit. **Product B** keeps falling in a straight line and never flattens. It does not have

product-market fit, however large its early numbers look.

Reading the shape of a retention curve

Curve shape	What it means
Almost zero drop-off (95 of 100 users still there in month 12)	A strong habit or core utility. The retention curve looks like a flat horizontal line.
Loses 80% in the first 2 months, then the remaining 20% stay for years	A steep cliff followed by a long, stable tail. Common in products with a clear core use case.
Flat for a few months, then drops, then picks up and stays flat	Resurrection: users return later, often after a new feature or re-engagement campaign.
Constant downward diagonal (50%, 25%, 10%, then zero)	The product never flattens. No product-market fit yet.

PRACTICAL TEST

Plot retention by weekly cohort. If the curve flattens at a healthy level, you have a base to grow from. If it does not flatten, fix retention before you spend on acquisition, because the leaking bucket will waste the spend.

16. Growth loops beyond the funnel

AARRR is a linear funnel. Linear input gives linear output, and growth stops when spend stops. A compounding growth loop feeds itself: active users trigger the acquisition and activation of the next cohort.

Loop	How it works	Examples	Health metric
Viral and collaboration	A user creates an artifact, shares it, and invitees experience the product's value and sign up for their own workspace	Figma, Slack, Miro, Notion	Viral coefficient K greater than 1.0; loop cycle time
Content and SEO (user-generated)	Active users post public questions or answers; search engines index the pages; new searchers land and sign up	Stack Overflow, Quora, Pinterest, Glassdoor	Crawl-to-index ratio; UGC per active user
Paid reinvestment	Capital acquires a user; the user converts; gross margin from lifetime value funds the next acquisition	Duolingo, Uber, Wealthfront	LTV:CAC above 3.0; CAC payback under 12 months

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

17. HEART framework

HEART (from Google) is a user-experience framework. It organises metrics around five questions. Use it when AARRR is too focused on the funnel and you need to measure the quality of the experience.

Dimension	Question	Cred example
Happiness	Are users happy with the product?	App store rating
Engagement	Are users using the product deeply?	# of users who paid at least one bill and used at least one reward
Adoption	Are users starting to use the product?	# of users who pay their first credit card bill

Retention	Are users coming back repeatedly?	# of users who pay 3 consecutive credit card bills
Task success	Are users achieving their goals?	# of users who avoid late payments on bills

HEART VS AARRR

AARRR tells you where users are in their lifecycle. HEART tells you whether the experience is good. Many teams use both: AARRR to track the funnel, HEART to judge quality per feature.

18. OKRs and the metric cascade

OBJECTIVES

Visionary goals or north stars that set a clear direction. They are significant, concrete, action-oriented and inspirational. Example (Zoom): provide video experience excellence that makes every meeting feel like in-person.

KEY RESULTS

Time-bound, measurable outcomes that track progress towards the objective. They are specific, aggressive yet realistic, measurable and verifiable. Example (Zoom): reduce meeting disconnections by 30% for all calls in FY26.

How OKRs plan the work

An objective sits at the team level. Key results (KR1, KR2, KR3) measure it. Initiatives deliver the key results. Each level has a scope: the objective is set by the higher team and the key results belong to the team.

Spotify example: from vision to sprint

Vision: unlock the potential of human creativity by giving artists the chance to live off their art and fans the chance to enjoy it.

Objective	Key result (team)	Initiative
Support 10,000 new artists launching their first album in FY26	Decrease average time to launch an album by 30% (creator tools team)	Launch an AI-based creator tools suite
Generate $5M in artist revenue in FY26	Diversify up to 10% of artist revenue (new initiatives team)	Launch virtual concerts
Reach 1B monthly active users in FY26	Get 100M new users (growth team)	Launch v2 of Spotify Wrapped
Reach 10B hours of monthly streaming in FY26	Increase average session length by 20% (recommendations team)	Improve the recommendation engine using a better understanding of song vibes

The execution bridge runs from a long-term North Star (total monthly listening hours), through input drivers (discovery breadth, personalisation depth, social-sharing velocity), to team-level OKRs (for example, the search team's goal of an immediate, skip-free magic track, with a key result to lift day-7 skip-free playback from 68% to 82%), and finally to bi-weekly sprints and experiments.

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

19. Analytics tools hierarchy

Think of analytics as a pyramid. Each layer depends on the one below it.

Layer (bottom to top)	What happens there
1. Data collection	Events and properties are captured reliably (basic requirement)
2. Data warehousing and pipelining	Data is stored and moved into a usable shape (basic requirement)

3. Data analysis	Questions are answered with segments, funnels and cohorts. **This is where most teams get stuck.** A product analytics tool helps most here.
4. Insight generation	Patterns become explanations and recommendations
5. Decision automation	Decisions run themselves, for example automatic rollbacks. Few teams do this.

The class recommended a Mixpanel tutorial for the analysis layer, and an external overview of the product analytics market.

20. Metrics cheat sheet

Business metrics

- **Revenue:** ARPU, lifetime value (LTV), annual recurring revenue (ARR), gross merchandise value (GMV), average order value (AOV)
- **Costs:** customer acquisition cost (CAC), cost of goods sold (COGS), marketing costs, fixed and variable costs
- **Profit:** gross margin, net margin, unit contribution

Product metrics

- **Acquisition:** app installs, downloads, sign-ups, unique visitors, bounce rate
- **Activation:** MAU / WAU / DAU, session count, conversion rate
- **Engagement:** engaged users (MEU / WEU / DEU), session length, feature usage
- **Retention:** week-over-week (WoW), month-over-month (MoM), week-4 retention, churn rate
- **Satisfaction:** CSAT, NPS, app store rating, support tickets, bugs

Feature metrics

- Adoption, discovery (first-time adopters), repeat adopters
- Usage funnel, usage heatmap, task completion
- **Efficacy:** task success rate, task failure rate
- **Efficiency:** time to complete, steps to complete, cognitive load

Engineering metrics

- **Latency:** page load time (PLT), time to interactive (TTI), visually complete time (VCT), cold and warm start time, end-to-end (E2E) action latency, perceived latency
- **Reliability:** E2E action reliability, API reliability, mean time to respond (MTTR), mean time between failures (MTBF), downtime, API throttling rate

PART 2 · HOW PRODUCT MANAGERS TRACK METRICS DAY TO DAY

21. Interview answer template

When an interviewer asks "How would you measure success for X?", answer in this order. It takes about two minutes and shows structured thinking.

1. **Users and stakeholders.** Name each user type and its JTBD.
2. **North Star.** One metric that captures value delivered to the main user.
3. **Ladder.** Two or three inputs (leading, controllable) and the outputs they drive (lagging, business).
4. **Business metric.** How the product supports revenue or the business model.
5. **Guardrails.** One counter-metric per primary metric, for example retention, uninstalls, latency or trust.
6. **Diagnostics.** How you would find the cause if the North Star moved (funnel steps, cohorts, segments).

Common mistakes to avoid

- Choosing a vanity metric (total sign-ups, total downloads) that does not reflect value.
- Reporting a metric without a time grain or cohort.
- Optimising one metric without a guardrail, which invites Goodhart's Law.
- Measuring outputs only. Leading inputs are what the team can act on.

- Using averages to judge retention. Always look at cohorts.
- Declaring product-market fit from acquisition numbers. Check whether the retention curve flattens.

MODULE SUMMARY

Define success for each stakeholder (JTBD). Pick a North Star. Build a ladder of leading inputs and lagging outputs. Add guardrails to every primary metric. Track the lifecycle with AARRR and quality with HEART. Check retention curves for product-market fit. Connect the metrics to OKRs so the team can act on them.

Source material: Week 6 Analytics and Metrics class (Product Management Launchpad). Full class slides are available as a separate PDF. The quick-notes PDF lists the key pointers for revision.

Module 7: Product Analytics and Metrics

Detailed supplement: Goodhart's Law, the 2×2 metric architecture, guardrails, growth loops, retention curves and the OKR cascade

Written by Varun Kumar, for SkillNavora · Draft

Official site: www.skillnavora.in

Prepared for Skillnavora — Learn. Practise. Implement. Grow.

MODULE 7 · HOW TO USE THIS SUPPLEMENT

Coverage map: what is new and what is already in the module

Read this supplement next to the main Module 7 handbook and the detailed PDF. **Extended** means the handbook has the topic and this PDF adds the deck's example. **New** means the handbook did not have it. Company examples come from the deck and are not verified; check them before you quote them.

Topic	Already in the module?	Status here
Ladder of metrics: input and output	Yes (ladder section)	Extended: the causal link and what you can control
Leading and lagging metrics	Yes	Extended: examples side by side
North Star metric	Yes	Extended: the 'one action that creates value' test
Goodhart's Law and metric gaming	Partly (one mention)	New: proxy displacement and three PM traps
2×2: controllability and timing	No	New: the four quadrants
Counter-metric and guardrail architecture	Yes	Extended: a table of focus areas, primary metrics and guardrails
AARRR and HEART	Yes	Not repeated here
Growth loops versus linear funnels	No	New: viral, content and other loop types
Retention curve shapes	Partly	New: four shapes and how to read them
Google Maps metrics architecture (activity)	No	New: a worked model for two stakeholders
OKR cascade from North Star to sprints	Partly (OKR section)	Extended: four levels with one worked chain

NOTE ON THE SOURCE

The deck's Goodhart's Law and guardrail slides include numeric targets such as '<2 percent' or '>=25 percent'. They are illustrative. Set your own targets from your own baseline.

NEW · METRIC GAMING

Goodhart's Law: when a measure becomes a target

The deck's version of the idea: **'When a measure becomes a target, it ceases to be a good measure.'** Three mechanisms cause the damage:

Mechanism	What happens
Proxy displacement	Teams confuse a proxy (clicks, raw daily active users) with real customer value (completing the core task).
Path of least resistance	Under sprint or OKR pressure, teams take the easiest way to move the number.
Compounding side-effects	Unmonitored dimensions decline quietly until trust, retention or unit economics fall.

Three PM traps (illustrative)

Trap	Example from the deck	Lesson
Notification trap	Five times more promotional push alerts raise daily active users by 8 percent, while uninstalls and muted alerts rise by 40 percent.	Watch the guardrail: uninstalls and mutes.
Signup friction trap	Pre-ticking a newsletter box lifts registrations, and then the new users unsubscribe.	Count the quality of signups, not only the number.
Click-farm trap	Acquisition targets reached with low-intent or bot traffic.	Pair acquisition with retention and fraud guardrails.

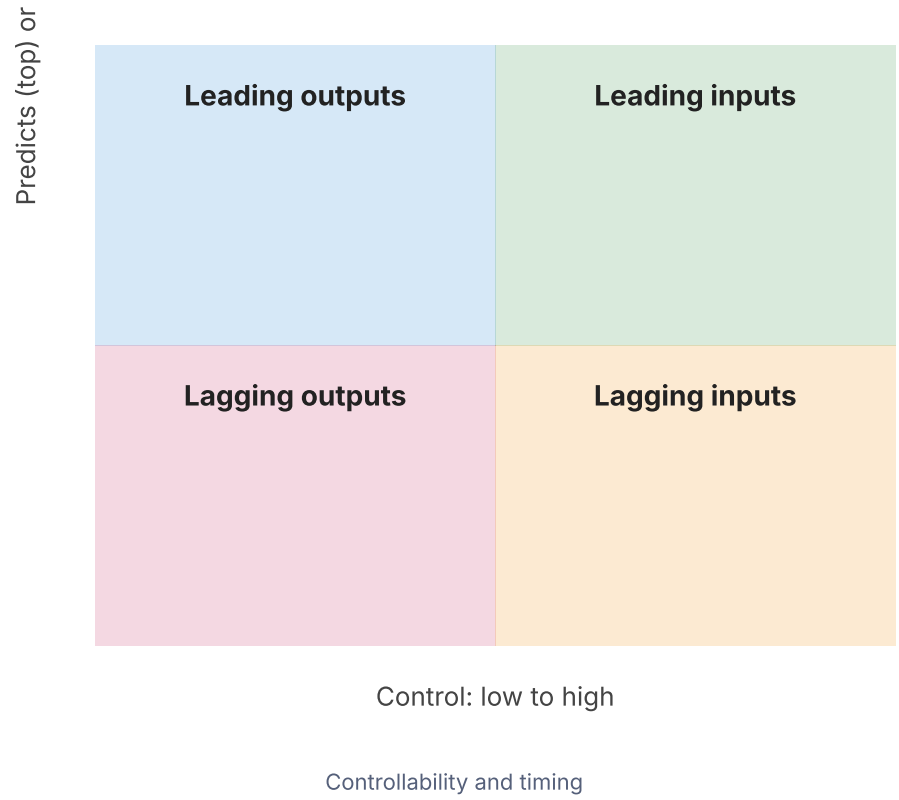
NEW · METRIC ARCHITECTURE

The 2×2: controllability versus timing

Sort every metric on two questions: can your team **control** it, and does it **predict** the future or **report** the past? That gives four quadrants.

Quadrant	What it means	Example from the deck (illustrative)
Leading inputs (high control, predictive)	The daily levers your team controls	Share of documents auto-verified on first submission (fintech); autocomplete latency (e-commerce)
Leading outputs (low control, predictive)	Early behaviour that signals long-term value	A team reaching a 'magic moment' in its first weeks; a user doing a key action on two devices
Lagging inputs	Activity you can control, but its results take time to show	Changes to onboarding flows measured on later cohorts
Lagging outputs (low control, reports the past)	Final results	Revenue; retained users at 90 days

Start with the leading inputs. They are what the team can change this week.



Controllability and timing

EXTENDED · GUARDRAILS

Primary metric, gaming risk and counter-guardrail

Every primary metric needs a guardrail: the number you refuse to damage while you push the first one. The deck's pattern:

Focus area	Primary metric (what we push)	Gaming risk (what can break)	Counter-guardrail (what we defend)
Acquisition	New signups and app installs	Low-intent or bot signups	Day-30 retention; fraud and chargeback rate
Engagement	Daily active users	Over-notifying users to create artificial activity	App uninstalls per 1,000 sessions; notification mute rate
Monetisation	Average revenue per user	Dark patterns and hidden renewals	Support ticket ratio; refund and dispute rate

Targets in the table are examples. Set your own from your baseline.

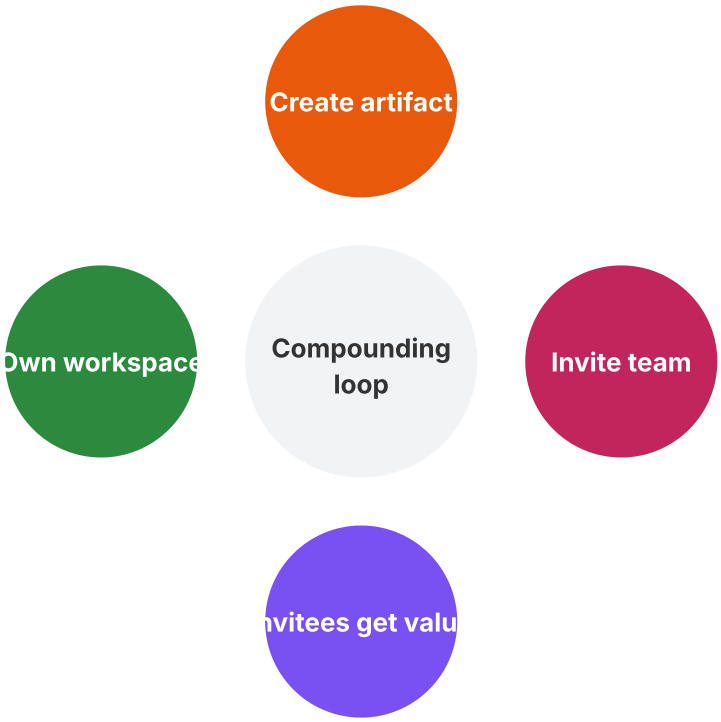
NEW · GROWTH

Linear funnels versus compounding loops

A linear funnel (acquisition, activation, retention, revenue) needs new input at every step. A loop feeds its own output back in: the actions of current users bring in and activate the next group.

Loop type	How it works (deck)	Metric to track
Viral or collaboration loop	A user creates an artifact, invites team members, and they start their own workspaces.	Viral coefficient above 1.0; loop cycle time
Content or search loop	An active user posts a question or review; the platform publishes a crawlable page; search engines index it; new visitors arrive and some join.	Pages indexed; share of new users from search

The deck names collaboration products such as Figma, Slack, Miro and Notion as examples of the viral loop. Check how each one works today before you use it as a case study.



A viral loop, from the deck

NEW · RETENTION CURVES

Four retention shapes and what they mean

Shape	What it looks like	What it tells you
Flat	Almost no drop-off; 95 of 100 users remain at month 12.	Strong habit or a core need; check the sample first.
Cliff then tail	Loses most users early, then the few who stay remain for years.	Fix onboarding and the first value moment.
Dip then recover	Flat for a few months, drops, then levels off again.	Find out what brought users back.
Steady decline	A constant downward line that never flattens; users are almost gone by month three.	No retained core; rethink the problem before growth spend.

Practice: for a product you use, draw the shape you expect, then check the real curve in analytics. Your guess is data about your own assumptions.

NEW · ACTIVITY

Google Maps metrics architecture

The deck's activity asks for metrics for two groups of stakeholders. Here is the model answer for drivers and commuters (illustrative):

Element	Answer
Core job to be done	Find the fastest, safest route with a reliable arrival time.
North Star metric	Share of trips taken on the best route (time, road condition, tolls, live traffic).
Leading metric	Number of navigation sessions started.
Lagging metric	Weekly active drivers; 90-day retention.
Input	Reroute time under two seconds; speed of hazard reports.
Output	Share of trips completed on the best route; trip satisfaction.
Counter-guardrails	Trip abandonment under 2 percent; arrival-time error under 3 minutes; distracted-driving alerts.

Then do the second group yourself: local businesses. Their job is to be found nearby and turn visits into paying customers. Write its North Star and guardrails.

EXTENDED · OKR CASCADE

From North Star to sprint

The deck shows four levels. Each level narrows the one above it. The example uses a music streaming company (the deck's figures are illustrative):

Level	What it is	Example from the deck
1. Company North Star	A 1 to 3 year metric	Total monthly listening hours
2. Input drivers	Levers assigned to teams for the year	Discovery breadth; personalisation depth; sharing
3. Team OKRs	Quarterly objective with measurable key results	Search team: make every search deliver a playable track; cut time to first song
4. Initiatives and sprints	The weekly work that moves the key results	Ship a faster search results screen

Key results must be measurable, time-bound and specific. 'Make search faster' is not a key result. 'Reduce time to first song from 4.2 seconds to 1.8 seconds this quarter' is.



From North Star to sprint

ASSIGNMENT

Build a metric architecture for a product

1. Pick a product you use. Name its users and their jobs to be done.
2. Write one North Star metric and explain why it captures the job.
3. Build a metric ladder with at least two leading metrics, two lagging metrics, and one input-to-output link.
4. Add a counter-guardrail for each primary metric, and say how it could be gamed.
5. Repeat the exercise for a second stakeholder group.

SELF-CHECK

Is each metric linked to a job? Did you name one way it could be gamed? Are your targets from a baseline, not invented?

Quiz with answers

1. State Goodhart's Law in the deck's words.
   Answer: 'When a measure becomes a target, it ceases to be a good measure.'
2. What are the two questions of the 2×2?
   Answer: can the team control the metric, and does it predict the future or report the past?
3. What is a counter-guardrail?
   Answer: a metric you refuse to damage while you push the primary metric.
4. How does a loop differ from a linear funnel?
   Answer: a loop feeds its output back in to bring in and activate the next group.
5. Which retention shape suggests a strong habit?
   Answer: the flat curve, after checking the sample.
6. Which level of the OKR cascade is a quarterly objective?
   Answer: level 3, team-level OKRs.

MODULE 7 · FINAL

Completion checklist

Completion checklist

Done	Item
☐	Read the coverage map and marked the new topics
☐	Named one metric-gaming risk for a product you use
☐	Placed five metrics on the 2×2
☐	Wrote guardrails for two primary metrics
☐	Built the metric architecture assignment
☐	Answered the quiz without notes

Analytics and Metrics: quick notes

DEFINITIONS

- **Metric:** a quantifiable measure of success (or failure) against a goal.
- **KPI:** the metrics that give the most insight into the goal.
- **Anatomy:** definition, value, time grain, cohort. Always state all four.
- **North Star Metric (NSM):** the one metric that signifies value delivered (for example Swiggy: successful orders delivered; Airbnb: bookings and successful stays).
- **Input metric:** something the team controls. **Output metric:** the result the input causes.
- **Leading metric:** an early signal of future success. **Lagging metric:** a confirmed past result.
- **Guardrail (counter) metric:** protects against side effects of the primary metric.

THE MENTAL MODEL (USE IT FOR ANY "DEFINE SUCCESS" QUESTION)

CUSTOMER

1. Identify all users and stakeholders.
2. Identify each Job To Be Done (JTBD).
3. Define a metric that captures JTBD completion.

BUSINESS

4. Identify the business goal (shareholder value, vision, mission).
5. Define a metric that shows the product supports it.
6. Add guardrails for each primary metric.

METRIC SYSTEM: FIVE LAYERS

Layer	Question	Example (B2B collaboration)
North Star	What long-term value are we creating?	Weekly active teams
Input	What behaviours drive the result?	Invites sent, workflows created
Output	What moved after launch?	Activation rate, workflow completion
Guardrail	What should not get worse?	Error rate, latency
Diagnostic	Where exactly is the problem?	Drop-off after the invite step

THE 2×2

	Predictive (leading)	Retrospective (lagging)
High control	Leading inputs: Slack invites sent in onboarding; CRM integrations on Day 0	Lagging inputs: story points shipped; paid budget deployed (effort, not value)
Low control	Leading outputs: Slack's 2,000-message magic moment; Duolingo day-7 streak	Lagging outputs: ARR, net dollar retention, LTV, LTV:CAC, EBITDA

Frameworks and checks

AARRR (LIFECYCLE)

Stage	Question	Sample metrics
Acquisition	Are users showing interest?	Installs, site visits, CAC
Activation	Are users reaching the magic moment?	Active users, first-value event rate
Retention	Are users coming back?	D/W/M retention, churn
Referral	Are users bringing others?	NPS, viral coefficient, referral share
Revenue	Are users paying?	Sales, LTV (LTV must be much greater than CAC)

HEART (QUALITY OF EXPERIENCE)

Happiness, **E**ngagement, **A**doption, **R**etention, **T**ask success. Use with AARRR: AARRR tracks the funnel, HEART judges quality.

RETENTION AND PRODUCT-MARKET FIT

- A healthy curve **flattens** at a sustainable level (product A in the class).
- A curve that keeps falling in a straight line never flattens: no product-market fit yet (product B).
- Fix retention before spending on acquisition. Otherwise the bucket keeps leaking.

GROWTH LOOPS (COMPOUNDING, NOT LINEAR)

- **Viral / collaboration:** viral coefficient K above 1.0 (Figma, Slack, Miro, Notion).
- **Content / SEO (user-generated):** crawl-to-index ratio and UGC per active user (Stack Overflow, Quora, Pinterest).
- **Paid reinvestment:** LTV:CAC above 3.0 and CAC payback under 12 months (Duolingo, Uber).

GOODHART'S LAW AND GUARDRAILS

- "When a measure becomes a target, it ceases to be a good measure."
- Traps: notification trap (DAU up, uninstalls up), signup friction trap (low-intent churn), feature tour trap (clicks without learning).
- Defences: paired counter-metrics, permanent holdout (for example 5%), qualitative triangulation.
- **Senior PM rule:** never ship an experiment without its paired guardrail.

OKRS

- **Objective:** visionary, significant, inspirational direction.
- **Key result:** time-bound, measurable, verifiable (for example, cut disconnections by 30% in FY26).
- Cascade: company North Star, input drivers, team OKRs, initiatives and sprints.

MODULE 7 · COMPLETE CLASS NOTES

Everything else from the class slides

Every point from the class slides that is not already explained earlier in this PDF, grouped by slide. Points already covered are not repeated.

Class · slide 2 Agenda

- Ladder of Metrics SkillNavora

Class · slide 5 From User Value to Business Value

- User job Get set up and go live faster Example: User outcomes B2B onboarding platform
- @ Time-to-value Business value metric Activation rate, expansion potential, lower churn risk $128K Revenue ;f' Business outcomes
- 0 Renewal confidence SkillNavora

Class · slide 6 What are Metrics

- The set of important metrics that provide insights are KPIs (Key Performance Indicators)
- Definition: What is it trying to measure.
- Time grain: The period over which the measurement is being made.
- Cohort: The slice or cohort for which the measurement is being made.

Class · slide 7 ●

- Measure of Physical health - Set of biomarkers (like HbA1c, Lipid panel, etc.)
- Measure of Social health - # of meaningful relationships
- Defining success metrics is hard when the goal is ambiguous!
- What are Metrics - Example SkillNavora

Class · slide 9 ●

- A product may have more than one type of user /stakeholder
- The higher order problem that the users want to have solved
- Device a success metric(s) that captures the completion of the JTBD
- The final goal for a business (for profit) is to generate shareholder wealth
- Companies also have a vision / mission which act as excellent goal statements
- Device a success metric(s) that captures support towards the business goal
- For example, how does the product support the business model of the company
- Guardrail metrics or counter metrics help guard against inadvertent effects SkillNavora Devise Devise

Class · slide 10 ●

- Content consumers: Total watch time or Total # of like, comment, share
- Revenue, # of new users added to IG, Drop in overall use of IG (guardrail) Exercise - Instagram Reels SkillNavora Devise Devise

Class · slide 11

SkillNavora

- SkillNavora Goodhart's Law & Metric Gaming Why single-metric optimization inevitably corrupts product outcomes "When a measure becomes a target, it ceases to be a good measure." - Marilyn Strathern (formalizing Charles Goodhart's economic principle for complex systems)
- The Gaming Mechanism Proxy Displacement Teams confuse the proxy metric (e.g. clicks or raw DAU) with true customer value (e.g. core workflow completion).
- Path of Least Resistance Under sprint or OKR target pressure, teams optimize the easiest way to move the number, not the hardest way to build real value.
- Real-World PM Traps The Notification Trap Boosting DAU by sending 5x promotional push alerts.
- Short-term DAU rises 8%; app uninstalls and muted alerts surge 40%.
- Yields low-intent churners and spam complaints.
- The Feature Tour Trap Forcing 5 unskippable modal walkthroughs to claim 100% feature adoption.
- Users click "Next" blindly without learning.
- The Senior PM Defense Paired Counter-Metrics Never evaluate an output metric in isolation; always pair volume with quality, retention, and friction guardrails.
- Holdout Groups & Audits Maintain permanent 5% holdout cohorts to verify true incrementality versus cannibalization of existing behavior.
- Qualitative Triangulation Pair metric dashboards with weekly customer interviews and support tickets to detect metric distortion early.

Class · slide 12

Ladder of Metrics (Categories or Type of Metrics)

- Ladder of Metrics (Categories or Type of Metrics) Input Metrics Output Metrics What can you control vs what you get Things you can control (Example: Number of user invites sent) Results you get (Example: Total Signup via user invitation) If inputs improve, outputs usually follow Input metrics measure the controllables you can directly influence (relatively speaking) Output metrics measure the outcomes of the input metrics.
- Input metrics have a causal effect on Output metrics SkillNavora

Class · slide 13

Leading Metrics

- Leading Metrics Lagging Metrics Predicting vs Reporting Early signals (future) (Example: Daily Active Users) Final results (past) (Example: Revenue) Leading tells you where you are going, lagging tells you where you reached Leading metrics are predictive and signal future success.
- Lagging metrics confirm what has already happened but are difficult to directly influence.

Class · slide 14

North Star Metric (NSM)

- North Star Metric (NSM) The One Metric That Matters Example: # of successful orders delivered for Swiggy Single metric / One Metric that Matters (OMTM) to signify success Ladder of Metrics (Categories or Type of Metrics) SkillNavora

Class · slide 15

Understanding the Ladder of Metrics

- Understanding the Ladder of Metrics Example North Star Metric (What is the one action that creates value) Number of bookings (& successful stays) Build the metric ladder Input (What we control) Output (What we get) Leading (Early signals, predicts the future) Lagging (Confirms the Past) Number of high-quality listings added # of views on listings Number of bookings made Revenue (commission per booking) SkillNavora

Class · slide 16

SkillNavora

- SkillNavora The 2×2 Metric Architecture: Controllability vs.
- Temporality Untangling Input vs.
- Output from Leading vs.
- Lagging metrics LEADING INPUTS (High Control, Predictive) THE DAILY LEVERS PRODUCT & GROWTH TEAMS CONTROL Slack: # of team member invites sent during initial onboarding SaaS CRM: # of active CRM integrations connected within Day 0 E-Commerce: Search autocomplete latency and query zero-result rate Fintech: % of KYC documents auto-verified on first submission High operational agency + direct causal link to downstream success.
- LAGGING INPUTS (High Control, Retrospective) HISTORICAL ACTIONS COMPLETED BY THE ORGANIZATION Engineering: Story points shipped, PRs merged, or sprint velocity in Q3 Growth: Total paid marketing budget deployed across ad channels Sales: Total enterprise demo meetings conducted by sales team Content: Number of help center articles published this quarter Measures organizational effort spent, NOT resulting customer value.
- LAGGING OUTPUTS (Low Control, Retrospective) THE ULTIMATE BUSINESS & FINANCIAL OUTCOMES Financial: Annual Recurring Revenue (ARR) & Net Dollar Retention (NDR) Retention: Annual Logo Churn Rate & Customer Lifetime Value (LTV) Efficiency: LTV : CAC ratio and gross contribution profit margin Market: Quarterly EBITDA & consolidated shareholder equity growth Unforgiving scoreboard of past execution.
- Cannot be directly operated on.

Class · slide 17

Understanding the Ladder of Metrics

- Understanding the Ladder of Metrics # of app installs Monthly Active Users Subscription Revenue Ad Revenue Other Revenue Total Listen Time # of Playlist creations Efficacy of Recommendation System # of Users who follow 10 artists in the first week (Magic Moment) Leading Metrics Lagging Metrics North Star Metric Example SkillNavora

Class · slide 23

Part 1: Define the below things for Google Maps

- Part 1: Define the below things for Google Maps
- Device a success metric(s) that captures support towards the business goal Part 2: Create the ladder of metrics
- Make sure you call out some leading metrics, some lagging metrics, and the input / output relationship between some metrics Activity Time SkillNavora Devise Devise

Class · slide 24 **SkillNavora**

- SkillNavora Activity Solution - Google Maps Metrics Architecture Deconstructing JTBD, North Star, Metric Ladder, and Guardrails across Stakeholders
- LAGGING Leading: # of active navigation sessions started Lagging: Weekly active drivers (WAU), 90-day app retention INPUT OUTPUT LINK Input: Reroute latency (<2s), hazard reporting speed Output: % completed optimal trips High trip NPS COUNTER GUARDRAILS Trip abandonment rate (<2%), ETA error variance (<3 min), distracted driving alerts
- Local Businesses CORE JOB TO BE DONE (JTBD) Drive organic discoverability and convert nearby foot traffic into verified paying customers.
- LAGGING Leading: # of claimed business profiles, updated menus Lagging: Merchant ad spend, merchant quarterly retention INPUT OUTPUT LINK Input: Profile completeness score, review response rate Output: Place sheet views In-store physical visits COUNTER GUARDRAILS Fake/duplicate listing rate (<0.1%), spam review volume, merchant dispute resolution latency
- NORTH STAR METRIC Net High-Intent Monetization (Local Search Ads + Google Maps Platform APIs).

Class · slide 26

- Metric Frameworks - AARRR SkillNavora

Class · slide 27

- Common acquisition channels include ads, SEO, content marketing, sales, affiliate marketing, community building, user referrals etc.
- Customer acquisition goes through the AIDA (Awareness, Interest, Desire, Action) funnel
- High acquisition (at least in the short term) does not require you to have achieved PMF
- A product with high acquisition but poor retention resembles a leaky bucket
- A steady growth in acquisition is needed to sustain a product since there will always be some users who leave

Class · slide 28

- An activated user is said to have gone through a "magic" moment
- If acquisition is un-targeted and broad, most users fail to get activated
- Products actively try to get their users to go through the magic moment
- For example, Twitter realized that once you followed 30 people you were more likely to come back so they suggest popular accounts when you sign up
- Dropbox saw that users who uploaded at least one file were much more likely to use Dropbox again and so they encourage you to upload a file during signup.
- Active Users AARRR - Activation SkillNavora

Class · slide 29

- Retention occurs when a customer is using the product repeatedly
- This requires the customer to perform the key value deriving activity repeatedly over a period of time
- Long term user retention is a good indicator of PMF
- Products that inculcate "loyalty" or "habits" among users generally have the highest retention
- Growing slowly gives you a better shot at retaining existing users
- Users that have churned can be reactivated

Class · slide 30

- Referral occurs when a customer recommends the product to other users
- Customers can refer the product to others through word of mouth
- Products that have an organic referral chain have low customer acquisition costs
- Products can encourage referral by building growth or viral loops in the product
- For example, discounts for referring friends, Products that sync with contact lists and send connection invites, Products creating shareable content that can be posted on social media
- Net Promoter Score (NPS), Viral coefficient, Customer Acquisition Costs (CAC) AARRR - Referral Book: Hooked: How to Build Habit-Forming Products - Nir Eyal SkillNavora

Class · slide 31

- Revenue occurs when a customer starts generating money for the product
- A product that has high number of retained and revenue providing users has achieved product market and business model fit
- Products need the right business model to generate sufficient revenue
- The customer LTV should be >> CAC for a sustainable business
- Sales, Lifetime Value (LTV) AARRR - Revenue SkillNavora

Class · slide 32

AARRR metrics for Cred (credit card payment)

- AARRR metrics for Cred (credit card payment)
- Activation - # of users who pay their first credit card bill or # of users who set up their credit cards on Cred
- Referral - # of users who have shared their Cred rewards on social media or # of users who successfully signed up using a referral code/link

Class · slide 33

Understanding product market fit

- Understanding product market fit SkillNavora

Class · slide 34

SkillNavora

- SkillNavora Beyond Linear Funnels: Compounding Growth Loops Why linear AARRR funnels leak capital while closed growth loops build enduring enterprise value LINEAR FUNNEL (AARRR): Acquisition Activation Retention Revenue.
- Linear input yields linear output.
- Money stops Growth stops.
- COMPOUNDING LOOP: Active cohort actions automatically trigger the acquisition and activation of the next cohort.
- Viral / Collab Loop PRODUCT-LED GROWTH
- User signs up & creates artifact (Doc, Board, Design)
- User shares / invites team members to collaborate
- Invited members experience immediate product utility
- Invited members register their own workspaces Exemplar Companies: Figma, Slack, Miro, Notion Viral Coefficient (K > 1.0) & Loop Cycle Time
- Content / SEO UGC Loop ORGANIC INDEXING ENGINE
- Active user posts a question, code snippet, or review
- Platform publishes public, highly crawlable page
- Google indexes page; captures high-intent long-tail search
- New searchers land on page, solve problem, and create accounts Exemplar Companies: StackOverflow, Quora, Pinterest, Glassdoor Crawl-to-Index Ratio & UGC Generation per Active User
- Paid Reinvestment Loop PERFORMANCE SCALER
- Deploy capital to acquire user via targeted ad
- User activates and converts to paying customer
- Customer delivers high Lifetime Value (LTV)
- Gross margin from LTV is reinvested to buy next user Exemplar Companies: Duolingo, Uber, Wealthfront LTV : CAC Ratio (>3.0) & CAC Payback Window (<12m)

Class · slide 35

Retention Curves

- If 100 people sign up, 95 are still there in Month 12.
- It looks like a straight horizontal line
- It loses 80% of users in the first 2 months.
- However, the 20% who remain stay for years.
- It stays flat for a few months, then drops significantly, but then picks up and stays flat in the later months
- It never flattens out.
- In Month 1 you have 50%, Month 2 you have 25%, Month 3 you have 10%, and eventually, it hits zero.
- Guess the product - Retention SkillNavora

Class · slide 36

- Metric Frameworks - HEART SkillNavora

Class · slide 37

HEART metrics for Cred (credit card payment)

- Engagement: # of users who have paid at least one credit card bill, and used at least one reward
- Task success: # of users who have avoided late payments on credit card bills HEART - Example SkillNavora

Class · slide 39

Hierarchy of Analytics

- Hierarchy of Analytics Tools Tutorial for Mixpanel More reading: https://www.news.aakashg.com/p/the-product-analytics-market-overview SkillNavora

Class · slide 40

Metrics Cheat Sheet

- Margin: Gross, Net, Unit Contribution Product metrics

Class · slide 41

Metrics Cheat Sheet

- Metrics Cheat Sheet Feature metrics
- Efficacy: Task success rates, Task failure rates
- E2E action perceived latency

Class · slide 44

Key Results

- Example (Zoom): Improve call stability by reducing meeting disconnections by 30% for all calls in FY26 OKRs - Definition Objectives

Class · slide 45

OKRs as a means to planning

- OKRs as a means to planning Objective KR1 KR2 KR3 Initiative 1 Initiative 2 Initiative 3 Initiative 4 Initiative 5 Objective Scope of a team Scope of the higher team SkillNavora

Class · slide 46

SkillNavora

- SkillNavora The Execution Bridge: North Star to OKR Cascade Translating long-term company vision into quarterly objectives and weekly engineering sprints LEVEL 1: COMPANY NORTH STAR METRIC (1 - 3 Year Strategic Horizon) Spotify North Star: Total Monthly Listening Hours Captures mutual value: listeners are deeply engaged; artists gain audience reach and streaming royalty payouts.
- LEVEL 2: INPUT DRIVERS / L1 METRICS (Annual Portfolio Resource Allocation) Driver 1: Discovery Breadth | Driver 2: Personalization Depth | Driver 3: Social Sharing Velocity Decomposes the North Star into distinct, mathematically correlated input levers assigned to specific PM pillars.
- LEVEL 3: TEAM-LEVEL OKRS (Quarterly Execution Focus & Accountability) Search Team Objective: Make every search query deliver an immediate, skip-free "Magic Track"
- KR 1: Increase Day-7 skip-free playback rate from 68% to 82%
- KR 2: Reduce time-to-first-song from 4.2s to 1.8s LEVEL 4: INITIATIVES & SPRINTS (Weekly Engineering, Design & Growth Backlog) Shipped in Bi-Weekly Sprints & Feature Experiments
- Sprint 1: Ship client-side vector search autocomplete
- Sprint 2: Pre-cache top 100 personalized tracks on app launch

Class · slide 47

Slide notes

- Spotify Vision: Unlock the potential of human creativity - by giving a million creative artists the opportunity to live off their art and billions of fans the opportunity to enjoy and be inspired by it
- KR3: Reach a MAU of 1B in FY26

Class · slide 48

- (Creator tools team) KR: Decrease average time taken to launch an album by 30% in FY26
- (New initiatives team) KR: Find ways to diversify upto 10% of artist revenue in FY26
- (Growth team) KR: Get 100M new users in FY26
- (Recommendations team) KR: Increase average session length by 20% in FY26
- I: Make improvements to recommendation engine through better understanding of song vibes OKRs - Example SkillNavora up to

MODULE 7 · SLIDE VISUALS

Visual slides from the class

These slides are mostly pictures and diagrams, so they are shown as images. Read them with the notes above.



Slide 1



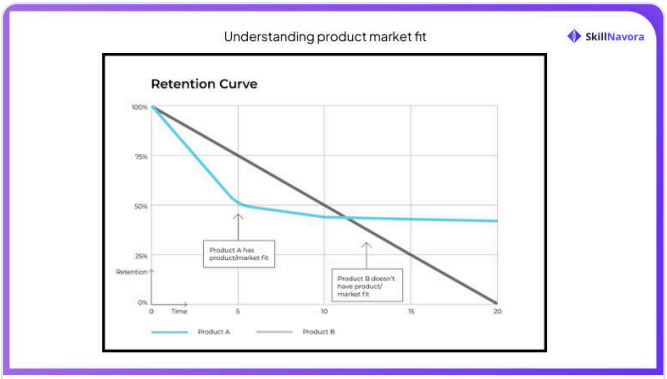
Slide 3



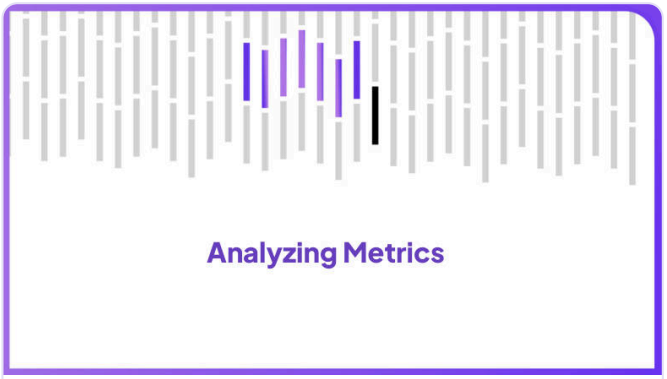
Slide 8



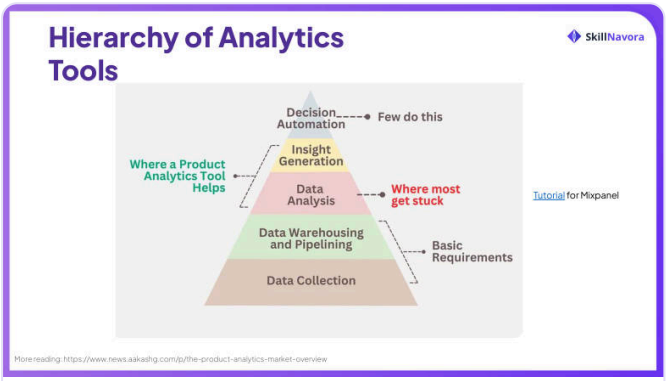
Slide 21



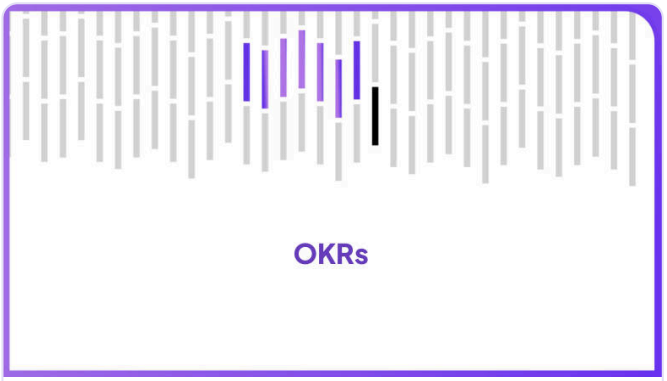
Slide 33



Slide 38



Slide 39



Slide 42



Slide 43